

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CE) Examination

November 2012

CE303 Computer Graphics

Date: 30.11.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following questions. [06]

1. Define the terms: (1) Horizontal Retrace (2) Aspect Ratio.
2. Explain DVST.
3. Differentiate Boundary-Fill algorithm and Flood-Fill algorithm.

(b) Explain Raster Scan System in detail. [05]

Q - 2 (a) Derive Bresenham's Line Drawing algorithm for $|m| < 1$ and write down the equations for $|m| > 1$. [04]

(b) Explain Scan Line Polygon Fill algorithm. What is the disadvantage of the algorithm? Write down the method to solve the disadvantage. [04]

(c) Answer the following questions. [04]

1. Explain Joystick and Data Glove in brief.
2. Explain odd-even rule to identify interior region of self-intersecting edges of polygon.

OR

Q - 2 (a) Two end-points of a line are: (3,2) and (8,9). Calculate intermediate pixel positions using DDA Line Drawing algorithm. [03]

(b) Answer the following questions. [05]

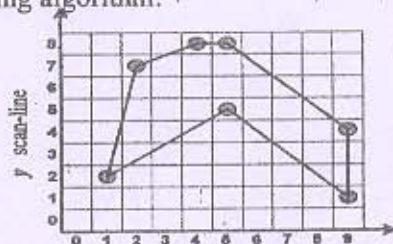
1. What is frame buffer? How long would it take to load a 1280 by 1024 frame buffer with 12 bits per pixel if transfer rate is 1Mbps? [02]
2. Which are the advantages and disadvantages of the DDA Line Drawing algorithm? [02]
3. Differentiate Emissive and Non emissive displays. [01]

(c) Calculate the pixel positions that are nearest to the original path of an ellipse having $r_x=10$ and $r_y=8$ centering at (0,0). Use Midpoint ellipse algorithm. [04]

Q - 3 (a) Explain Refresh Cathode-Ray Tubes with all components in detail. [04]

(b) Derive Midpoint circle drawing algorithm for the octant where slope of the curve varies between 0 to -1. [04]

(c) For a given polygon, prepare the Edge Table (ET) and Active Edge Table (AET) used in scan-fill polygon filling algorithm. [04]



OR

- (c) Explain Boundary-Fill algorithm in detail.
What is 4-Connected method? What is the disadvantage of it?
How the disadvantage of 4-Connected method is resolved?

[04]

SECTION - II

- Q - 4 (a) What is the problem with Sutherland-Hodgeman polygon clipping? Explain Weiler-Atherton Polygon clipping algorithm with example. [04]
(b) Do as Directed.
1. Enumerate the steps to display the scene described in world coordinates to device coordinates. [07]
2. Magnify the triangle A(0,0), B(2,2), and C(10,4), to three times its size while keeping R(10,4) as fixed point. [03]
- Q - 5 (a) Prove that the multiplication of 2D transformation matrices for each of the following sequence of operations is commutative. [05]
1. Two successive rotations.
2. Two successive translations.
(b) Do as Directed.
1. Two end-points of a line: A(40,10), and B(110, 30). Rectangle Clip window is (5, 5) and (100, 50). Clip the line using Liang-Barsky Line Clipping algorithm. [07]
2. What are the disadvantages of Cohen-Sutherland Line Clipping algorithm and compare it with Nicholl-Lee-Nicholl Line Clipping algorithm. [04]
- OR
- Q - 5 (a) What is 2D shear transformation? Convert the unit square to shifted parallelogram using x-direction shear transformation operation where parameter shx = 1/2 and Yref = -1 and unit square dimensions are (0, 0), (1, 0), (0, 1) and (1, 1). [04]
(b) Answer the following questions.
1. Explain the term region codes. Write the steps of Cohen-Sutherland line clipping algorithm. [08]
2. Explain reflection with respect to any plane in 3D transformations. [05]
- Q - 6 (a) Explain Nicholl-Lee-Nicholl Line Clipping algorithm. [03]
(b) Prove that transformation matrix for the rotation about an arbitrary axis can be expressed as the composition of following seven individual transformation. [06]
$$R(\theta) = T^{-1} \cdot R_x^{-1}(\alpha) \cdot R_y^{-1}(\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \cdot T$$
- OR
- Q - 6 (a) How can we rotate an object with respect to any arbitrary line in 3-D? Derive matrices. [06]
(b) Explain RGB Model in detail. Which are the applications of it. [06]
